Chapter 10

Taxation and Regulation

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 9 used a tax as an INSTRUMENT. It never asked who hands over the money.

CHAPTER 7 GAVE US THE TOOLS

CONSUMER SURPLUS, PRODUCER SURPLUS and DEADWEIGHT LOSS. Every area on today's diagrams is one of those three.

CHAPTER 9 SET A TAX EQUAL TO HARM

A PIGOUVIAN TAX of exactly the MARGINAL EXTERNAL COST moves the market to the social optimum. That was the tax's SIZE.

TODAY: WHO BEARS IT

TAX INCIDENCE is a different question from the tax rate. The law names who writes the cheque. It does not decide who is poorer.

STATUTORY INCIDENCE is who the law makes liable for the tax. ECONOMIC INCIDENCE is whose income falls. They are almost never the same people.

Three Questions

A TAX and a PRICE CONTROL are both WEDGES: they stop trades that both sides wanted to make.



Who bears the burden of a tax



Does it matter who the law taxes?



What decides the split



What price controls do



Where government itself goes wrong

Every diagram today has the same shape: a WEDGE between what buyers pay and what sellers keep, and a quantity that falls.

PART 1

Who Actually Pays a Tax

The Law Names a Payer. It Does Not Choose One.



A $2 tax on every plate of jambalaya. The restaurant remits it. So the restaurant pays it — yes?

TAX INCIDENCE — who actually bears the burden of a tax, measured by whose income falls, not by whose name is on the payment.

STATUTORY INCIDENCE is a legal fact: the statute names a remitter. ECONOMIC INCIDENCE is an economic outcome, and the market settles it.

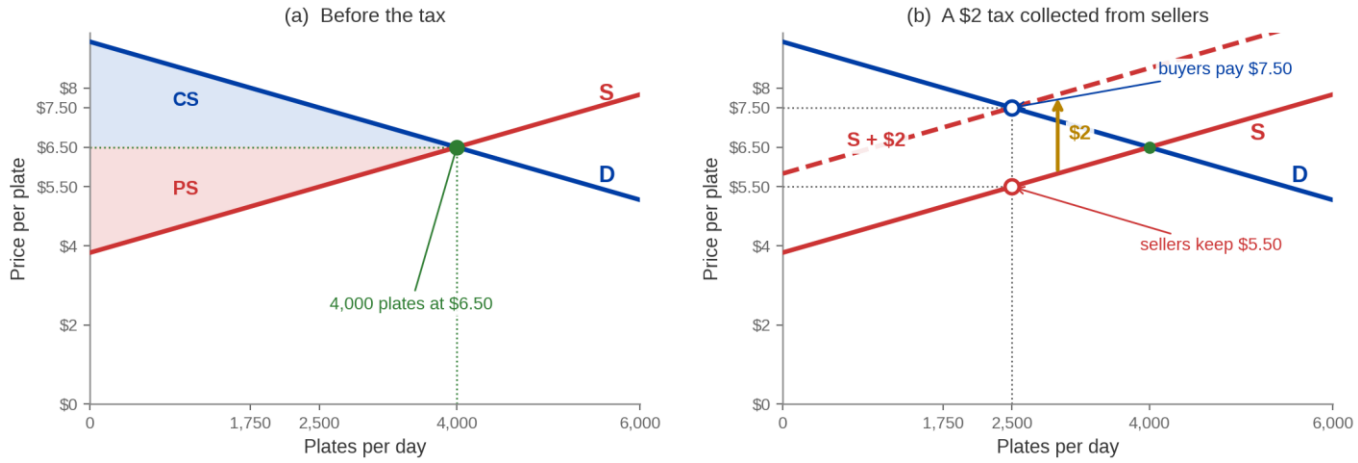
The restaurant will try to pass the tax on by raising the menu price. Whether it can, and by how much, is what the next four pages work out — and the answer turns out to have nothing to do with the wording of the law.

The question is not WHO REMITS. The question is WHOSE PRICE CHANGES — and both prices change.

Image: Acemoglu, Laibson & List, Microeconomics 3e

A Tax on Sellers Shifts Supply Up by Exactly the Tax

The market clears at 4,000 plates and $6.50. A $2 TAX raises every seller's required price by $2.

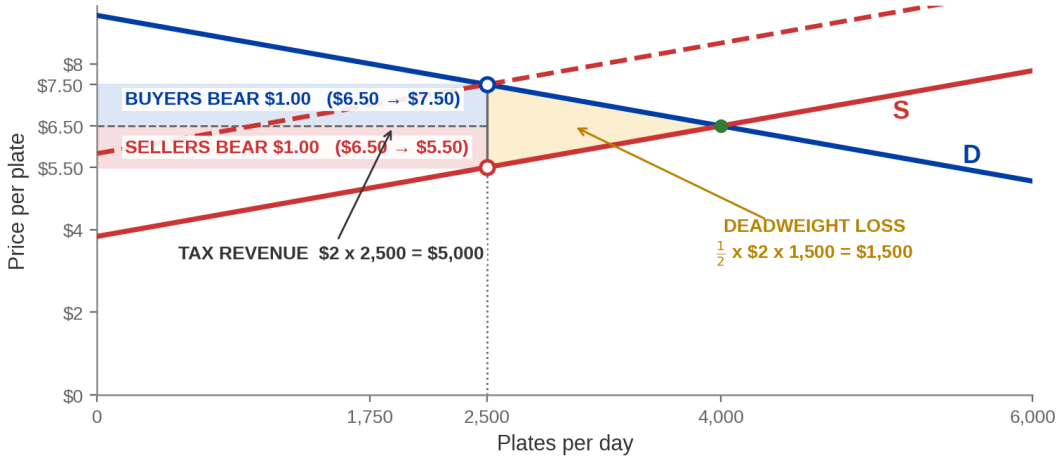


Quantity falls 4,000 → 2,500. Buyers pay $7.50 and sellers keep $5.50 — the TAX WEDGE is $2 and both prices move.

Market drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 10

Where the $2 Goes, and the Part Nobody Gets

TAX REVENUE = tax × post-tax quantity. DEADWEIGHT LOSS = the trades the wedge prevented.

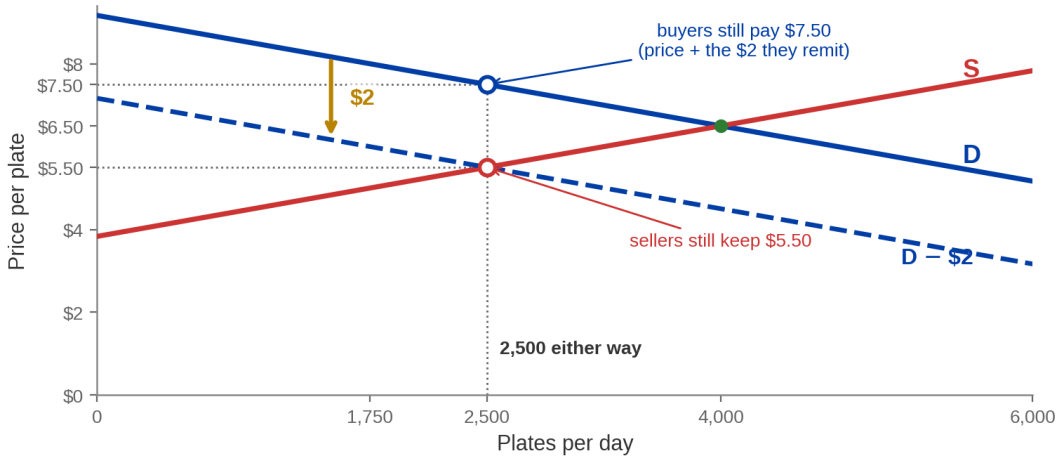


TAX REVENUE $5,000 a day. DEADWEIGHT LOSS $1,500 a day — value destroyed, not transferred, and nobody collects it.

Market drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 10

The Same $2, Collected From Buyers Instead

A tax on buyers shifts DEMAND down by the tax — what sellers may charge falls by $2 at every quantity.



2,500 plates. Buyers pay $7.50. Sellers keep $5.50 — identical. The TAX WEDGE does not care which side of it the law stands on.

Market drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 10

Two Taxes, One Outcome

STATUTORY INCIDENCE is a legal fact. ECONOMIC INCIDENCE is a market outcome.

	Tax collected from SELLERS	Tax collected from BUYERS
Which curve moves	Supply, UP by $2	Demand, DOWN by $2
Quantity traded	2,500 plates	2,500 plates
Buyers pay	$7.50	$7.50
Sellers keep	$5.50	$5.50
Tax revenue	$5,000 a day	$5,000 a day
Deadweight loss	$1,500 a day	$1,500 a day

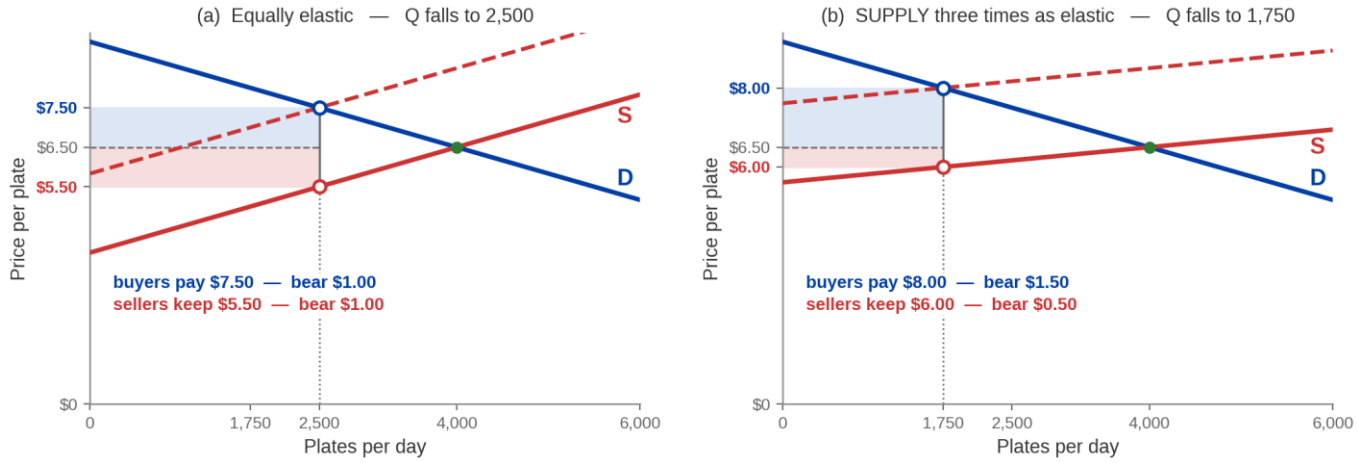
Read DOWN either column and the answer is the same. A $2 WEDGE is a $2 WEDGE, whoever the statute names.

PART 2

What Decides the Split

Make Supply More Elastic and Buyers Pay More

Sellers who can walk away easily do. The side that can ESCAPE the tax escapes it, and the other side absorbs it.

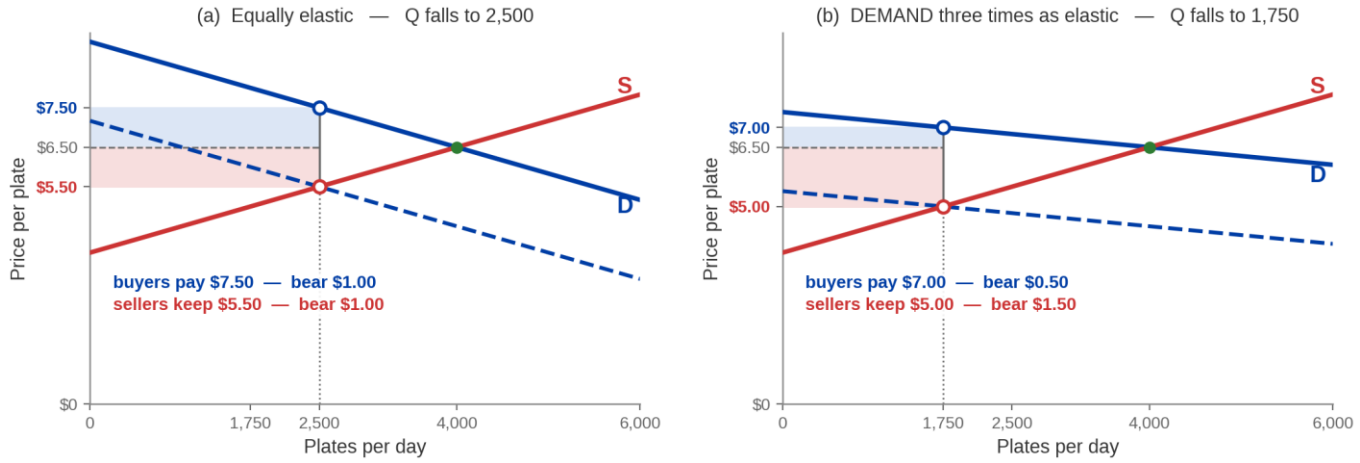


ELASTIC SUPPLY → burden on BUYERS: $1.50 of the $2, against $0.50 for sellers.

Panel (b) drawn holding DEMAND genuinely constant — see speaker notes

Make Demand More Elastic and Sellers Pay More

Now BUYERS are the side with somewhere else to go — and the ELASTICITY of demand turns the burden right round.

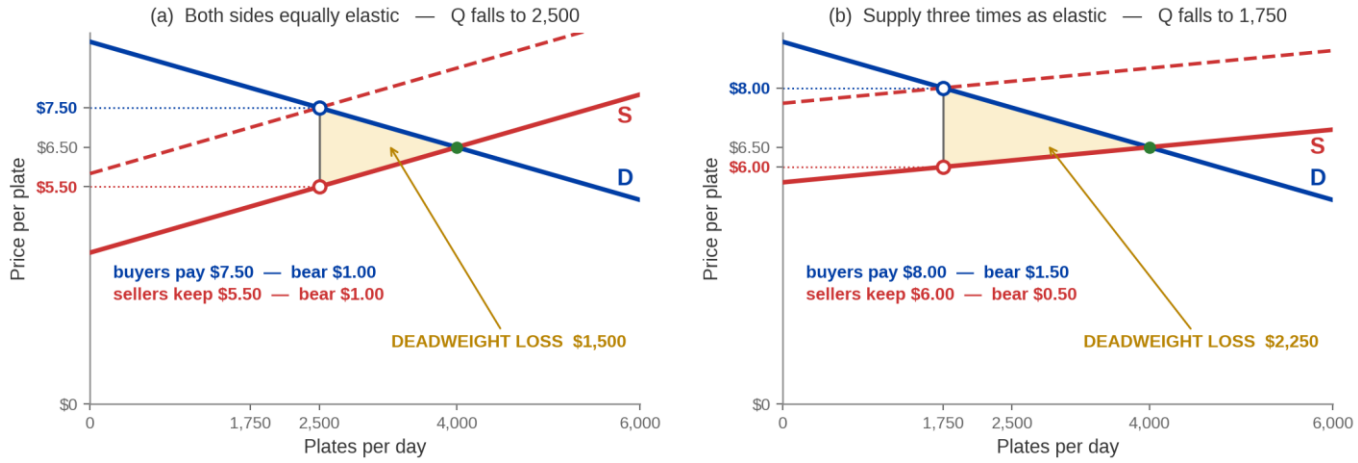


THE RULE: the burden falls on the LESS ELASTIC side. Elastic demand → sellers bear $1.50 of the $2.

Panel (b) drawn holding SUPPLY genuinely constant — see speaker notes

The More Elastic the Market, the More the Tax Destroys

Elasticity does two things at once: it moves the BURDEN, and it enlarges the DEADWEIGHT LOSS.



Same $2 tax, same revenue per plate — DEADWEIGHT LOSS rises $1,500 → $2,250 because more trades disappear.

PART 3

Regulation, and Where It Goes Wrong

Two Ways to Intervene Without Taxing

REGULATION — government action aimed at the quantity, price, quality or safety of what is traded.

DIRECT REGULATION

Also called COMMAND-AND-CONTROL: the government sets the AMOUNT of an activity rather than its price. Emissions caps, drug approval, vehicle standards, licensing.



PRICE CONTROLS

The government sets the PRICE instead.

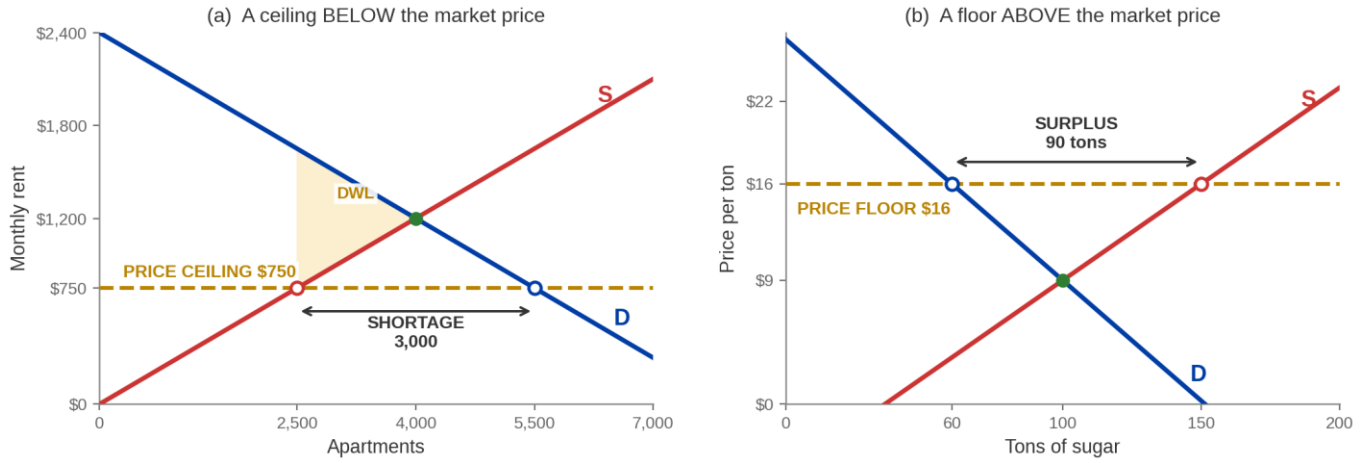




A tax changes the PRICE and lets the market pick the quantity. DIRECT REGULATION picks the quantity. A PRICE CONTROL picks the price and lets neither clear.

A Ceiling Makes a Shortage. A Floor Makes a Surplus.

A PRICE CEILING bites BELOW the market price; a PRICE FLOOR bites ABOVE it. Either way the SHORT SIDE decides what trades.

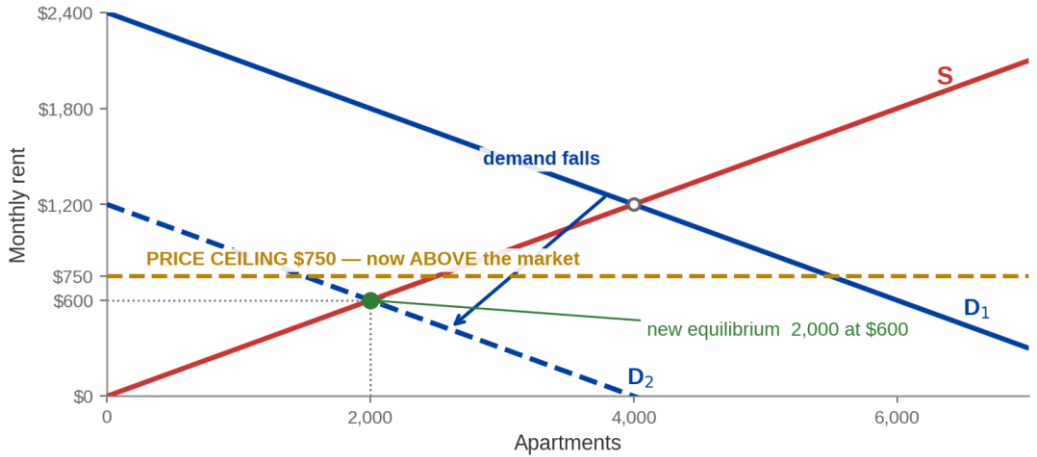


Rent capped at $750: 5,500 wanted, 2,500 offered, SHORTAGE 3,000. Sugar floored at $16: 150 offered, 60 wanted, SURPLUS 90.

Markets drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 10

A Ceiling Only Bites If the Market Is Above It

The regulation did not change. The MARKET moved under it — and the CEILING stopped doing anything.



Demand falls, rent would clear at $600, and a $750 ceiling is now ABOVE the market price: NO SHORTAGE, no DEADWEIGHT LOSS, no effect at all.

Market drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 10

Government Failures

GOVERNMENT FAILURE — inefficiency caused by the government's own intervention.

BUREAUCRATIC COST

More rules need more people to enforce them, and those people had another use. That is an OPPORTUNITY COST, not an accounting entry — and it is a reason to weigh a regulation, not a reason against having one.

CORRUPTION

Public funds diverted to private use. Not confined to government: rent control produces landlords demanding payments under the table, which is corruption generated in the PRIVATE sector by a public rule.

THE UNDERGROUND ECONOMY

Where a control creates a shortage or a tax makes an activity expensive, trade moves off the books. BLACK MARKETS grow exactly where the wedge is biggest — the shortage from the previous page has to go somewhere.

DEADWEIGHT LOSS is the cost the diagram shows. GOVERNMENT FAILURE is the cost it does not — and both belong on the same side of the ledger.

Chapter 10: The Full Picture

	What it says	Key terms	Where you saw it
Who bears a tax	The law names a remitter; the market decides who is poorer. Both prices move.	TAX INCIDENCE, STATUTORY vs ECONOMIC	§10.1.3 · the $2 jambalaya tax
It does not matter who is taxed	Tax the seller or tax the buyer: same quantity, same two prices, same revenue.	TAX WEDGE	§10.1.3 · the two-column table
What decides the split	The burden falls on the LESS elastic side — whoever cannot walk away.	ELASTICITY, INCIDENCE	§10.1.3 · the two elasticity figures
What a tax costs	Revenue is transferred; the triangle is destroyed. Both grow with elasticity.	DEADWEIGHT LOSS, TRANSFER	§10.1.3 · ½ × tax × fall in Q
Regulation	Direct limits, or price controls that bind only on one side of the market.	COMMAND-AND-CONTROL, CEILING, FLOOR, BINDING	§10.2 · rent and sugar
Government failure	Bureaucracy, corruption and black markets are costs no supply diagram shows.	GOVERNMENT FAILURE	§10.3

Not lectured, and in your reading: §10.1.1 and §10.1.2 (what government taxes and spends, progressive / proportional / regressive, average vs marginal tax rate), §10.4 (equity vs efficiency) and §10.5 (consumer sovereignty and paternalism).

KEY TAKEAWAYS

Chapter 10: Taxation and Regulation

- TAX INCIDENCE — the law names who remits a tax; the market decides whose income falls. They are rarely the same.
- THE TAX WEDGE — tax the buyer or tax the seller and every number is identical: same quantity, same two prices, same revenue.
- ELASTICITY DECIDES — the burden falls on the side that cannot walk away, and the more elastic the market the more trade the tax destroys.
- DEADWEIGHT LOSS — revenue is transferred, the triangle is destroyed: ½ × tax × the FALL in quantity.
- BINDING OR NOT — a CEILING acts only below the market price and a FLOOR only above it, and GOVERNMENT FAILURE is the cost no diagram shows.